Lecture 13:
Reproducible Research + Project Info

IS 618: Social Media Data Analysis
Indira Sen, FSS 2026

Recap: Ethical Considerations

What is Ethics?

- Ethics: “the discipline concerned with what is morally good and bad and morally right and wrong. The term is also applied to any system or theory of moral values or principles.” [Britannica]

- What should you take away from this lecture:
    - When analyzing social media data for research or beyond, how do you ensure that your work is ethical and does no harm?
    - How do you think about the ethical impacts of your projects?

Ethical Principles to Questions

- **Respect for Persons:** treating people as autonomous and honoring their wishes.
- **Beneficence:** understanding and improving the risk/benefit profile of your study, and then deciding if it strikes the right balance.
- **Justice:** ensuring that the risks and benefits of research are distributed fairly.
- **Respect for Law and Public Interest:** include all relevant stakeholders, not just research participants

Do you treat all research subjects with respect and dignity?

How do you minimize risks and maximize benefits?

Which people benefit from your research? Does it make the powerful more powerful?

What is the impact on people beyond your research subjects, e.g., the general public?

Putting Ethics into Action: Responsible AI and Science

- For research, reproducibility is paramount
- But sometimes, conflict with privacy, especially with sensitive subjects
- Also conflicts with dual use — malicious usage
- Recommendations*
  - Share your code
  - Share your data

* after considering costs and benefits

NeurIPS 2021 Paper Checklist Guidelines

The NeurIPS Paper Checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. For each question in the checklist:

- You should answer yes, no, or n/a.
- You should reference the section(s) of the paper that provide support for your answer.
- You can also optionally write a short (1–2 sentence) justification of your answer.

You are encouraged to take a look at the checklist early on as it likely will

Reviewers will be asked to use the checklist as one of the factors in their e ency: "Did you include [information]?" While "yes" is generally preferable to "no", it is perfectly acceptable to answer "no" provided a prope it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). **In general, answering "no" or "n/a" is not grounds for rejection.** While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgement and write a justification to elaborate. All supporting evidence can appear either in the **main paper or the supplemental material**.

We provide guidance on how to answer each question below. You may additionally refer to our blog post introducing the checklist to learn more about its motivation and how it was created.

1. For all authors...
   - (a) Do the main clai
     - Claims in the pa
     - The paper's con long as it is clea
   - (b) Have you read th
     - Please read the

https://neurips.cc/Conferences/2021/PaperInformation/PaperChecklist

'Just What do You Think You're Doing, Dave?'* A Checklist for Responsible Data Use in NLP

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Abstract

A key part of the NLP ethics movement is responsible use of data, but exactly what that means or how it can be best achieved remain unclear. This position paper discusses the core legal and ethical principles for collection and sharing of textual data, and the tensions between them. We propose a potential

contribute to the development of a consistent standard for data (re-)use, embraced across

(Mulligan et al., 2019; Xiang and Raji, 2019) and "bias" (Blodgett et al., 2020). The result is differing expectations and standards, which makes it harder for fellow researchers to understand the terms of (re-)use of new resources and models, and also complicates peer review.

Experts in biomedical data research have concluded that "truly informed consent ... requires ormational nce hat hers" (Mittelstadt, 2019). The AI community has already started to work on the latter (Mantelero, 2018;

Rogers, Anna, Timothy Baldwin, and Kobi Leins. "Just What Do You Think You're Doing, Dave? A Checklist for Responsible Data Use in NLP." *Findings of the Association for Computational Linguistics: EMNLP 2021.*

Reproducible Research Pipeline

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

https://book.the-turing-way.org/reproducible-research/overview/overview-definitions

Why is reproducibility and replicability important?

- Cornerstone of good science: reproducibility of empirical results is an essential part of the scientific method

- Ensure trustworthiness, rigor, and transparency of your findings

- Selfish reasons:
  - Better for you communicate what you did
  - Prevent mistakes:
    - Science is **iterative** and **self-correcting** —> honest mistakes happen
    - Providing reproducible materials out means that others can critically analyze and correct mistakes

A Beginner's Guide to Conducting Reproducible Research

Reasons behind the reproducibility crisis

- Fraud
- Technical debt:
  - "There's a deadline, I'll do this later"
  - "It's just exploratory right now…"
- Lack of awareness of good practices
- Lack of incentives

DATA COLADA

Thinking about evidence, and vice versa

HOME TABLE OF CONTENTS FEEDBACK POLICY SEMINAR

[109] Data Falsificada (Part 1): "Clusterfake"

Posted on June 17, 2023 by Uri, Joe, & Leif

This is the introduction to a four-part series of posts detailing evidence of fraud in four academic papers co-authored by Harvard Business School Professor Francesca Gino.

In 2021, we and a team of anonymous researchers examined a number of studies co-authored by Gino, because we had concerns that they contained fraudulent data. We discovered evidence of fraud in papers spanning over a decade, including papers published quite recently (in 2020).

https://datacolada.org/109

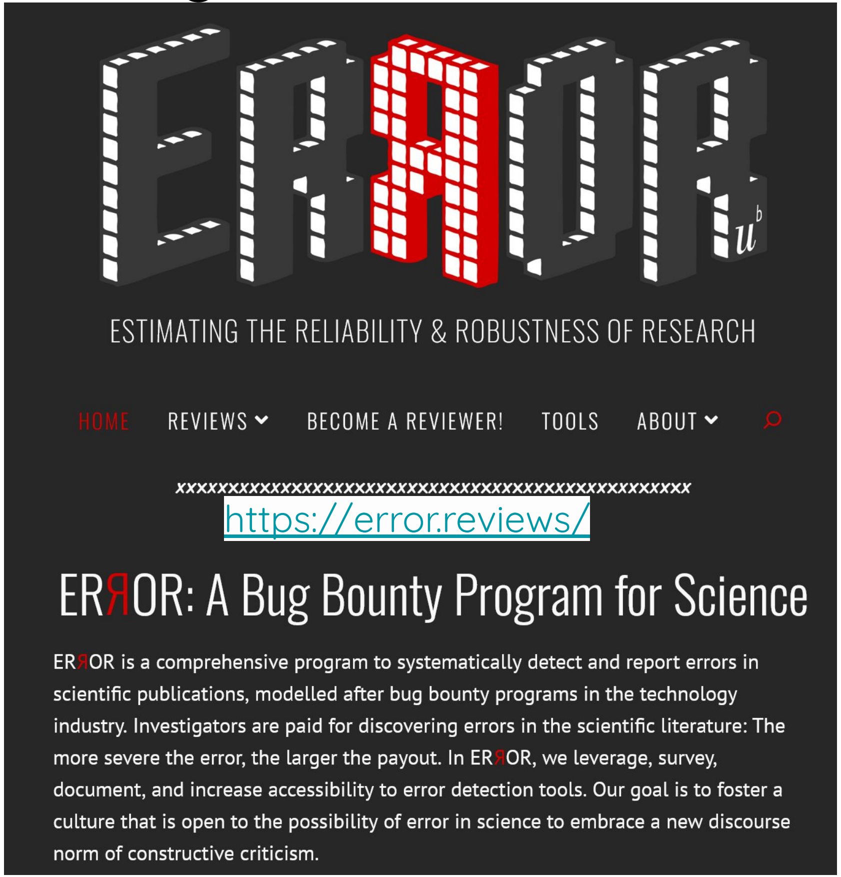
Reasons behind the reproducibility crisis

- Fraud

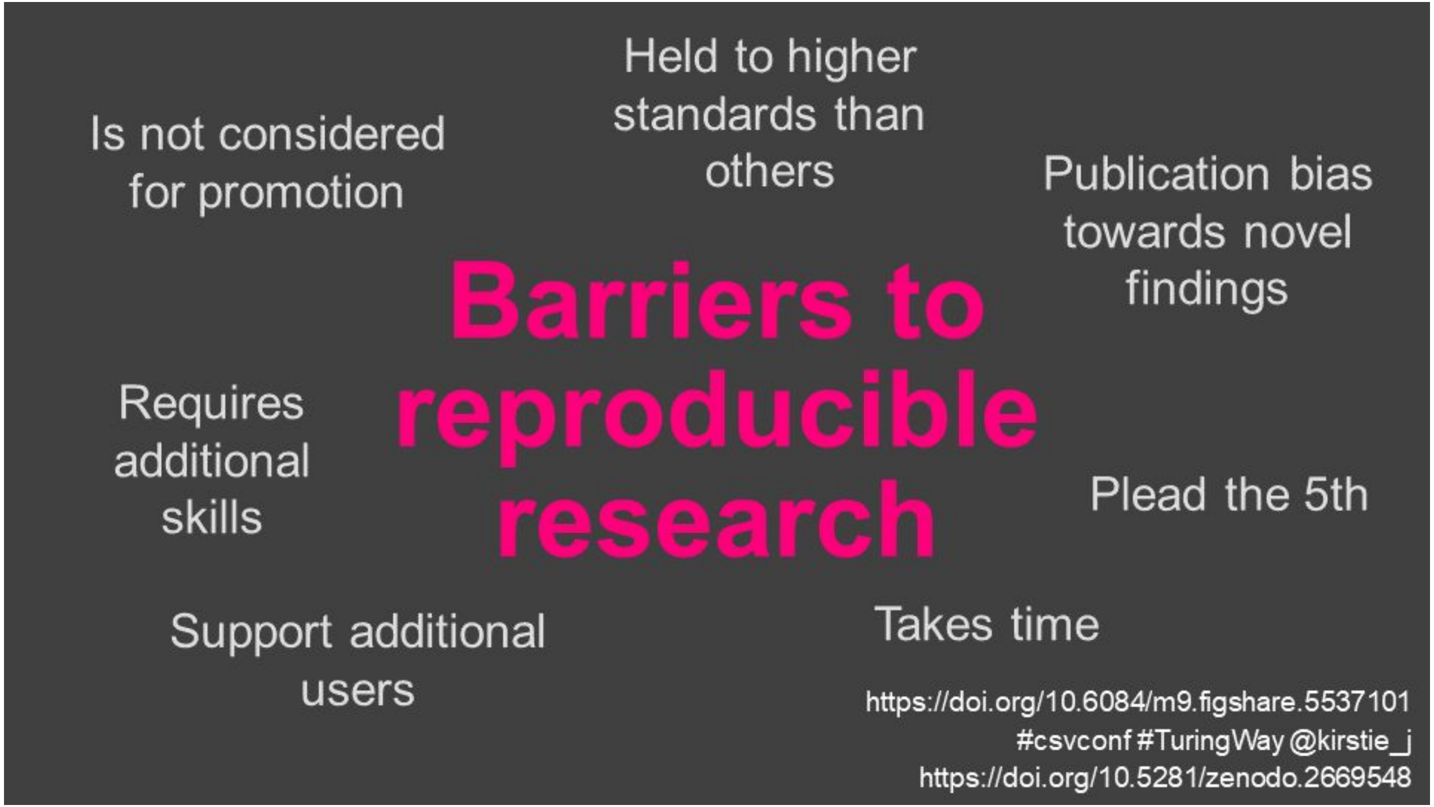
- Technical debt:
  - “There’s a deadline, I’ll do this later”
  - “It’s just exploratory right now…”

- Lack of awareness of good practices

- Lack of incentives



Common barriers to reproducibility



Common barriers to reproducibility

- Data
  - Inaccessible data
  - Poor or insufficient documentation
- Proprietary analysis tools, e.g., models or software
- Lack of reliability measures
  - **TIP:** always include error bars, confidence intervals
- Underpowered experiments
  - **TIP:** try to do a power analysis if you already have some exploratory results

Common barriers to reproducibility for CSS research

- Social media data sharing
  - Dataset decay
  - Tensions with welfare, privacy, and informed consent
  - Secondary data brokers providing little information about **data provenance**

Dataset	Lang	Size	Avail	Construct	Sharing	Hosting
Waseem2016 (Waseem and Hovy 2016)	EN	6,909	91/54%	HS	IDs	GitHub
WaseemHovy2016 (Waseem and Hovy 2016)	EN	16,791	64/50%	HS (sexism, racism)	IDs	GitHub
BenevolentSexism (Jha and Mamidi 2017)	EN	7,205	-/33%	HS (sexism)	IDs	GitHub
Davidson2017 (Davidson et al. 2017)	EN	25,296	•	HS & OL	Tweets	GitHub
Golbeck2017 (Golbeck et al. 2017)	EN	35,000	•	Harassment	Tweets on req	SH
Ross2017 (Ross et al. 2016)	GER	477	•	HS (immigrants)	Tweets	GitHub
Bohra2018 (Bohra et al. 2018)	EN,HI	4,067	69/69%	HS	IDs	GitHub
ElSherief2018 (ElSherief et al. 2018)	EN	28,498	-/32%	HS	IDs	GitHub
Founta2018 (Founta et al. 2018)	EN,HI	79,894	61/39%	HS	IDs / Tweets on req	SH / Zenodo
GermEval2018 (Wiegand, Siegel, and Ruppenhofer 2018)	EN,HI	8,541	•	OL	Tweets	GitHub
IberEval2018 (Fersini, Rosso, and Anzovino 2018)	EN	3,977	•	Sexism, misogyny	Tweets on req	SH
Rezvan2018 (Rezvan et al. 2018)	EN/HI	24,189	•	Harassment	Tweets on req	SH
Ribeiro2018 (Ribeiro et al. 2018)	EN	4,972	•	HS	Network(req)	GitHub / Kaggle
HASOC19 (Mandl et al. 2020)	EN,HI,GER	7,005	•	HS & OL	Tweets	SH
HatEval (Basile et al. 2019)	EN,SP	13,000	•	HS	Tweets	SH
OLID (Zampieri et al. 2019)	EN	14,100	•	OL	Tweets	SH
Ousidhoum2019 (Ousidhoum et al. 2019)	EN,AR	5,647	•	HS	Tweets	GitHub
Toosi2019 (Toosi 2019)	EN	31,961	•	HS sentiment	Tweets	Kaggle
ALONE (Wijesiriwardene et al. 2020)	EN	688	•	HS/OL (Toxicity)	Tweets on req	SH
Gomez2020 (Gomez et al. 2020)	EN	149,823	48/44%	Multimodal HS	Tweets, IDs, Img	SH
MeTooMA (Gautam et al. 2020)	EN	9,973	78/78%	HS	IDs	Dataverse
CMSB (Samory et al. 2021)	EN	2,743	87/85%	Sexism	Tweets on req	Datorium
Covid2021 (Wich, Räther, and Groh 2021)						
SWAD						GitHub

[The End of the Rehydration Era The Problem of Sharing Harmful Twitter Research Data]

Common barriers to reproducibility for CSS research

- Social media data sharing
  - Dataset decay
  - Tensions with welfare, privacy, and informed consent
  - Secondary data brokers providing little information about **data provenance**

- Use of black-box services
  - Little information about service development (e.g., ChatGPT, Perspective API, Botometer)

On the Challenges of Using Black-Box APIs for Toxicity Evaluation in Research

On the Challenges of Using Black-Box APIs for Toxicity Evaluation in Research

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Abstract

Perception of toxicity evolves over time and often differs between geographies and cultural backgrounds. Similarly, black-box commer-

Automatic toxicity detection tools, which often use machine learning algorithms to quickly analyze large amounts of data and identify patterns of toxic language, are a popular and cost-effective method

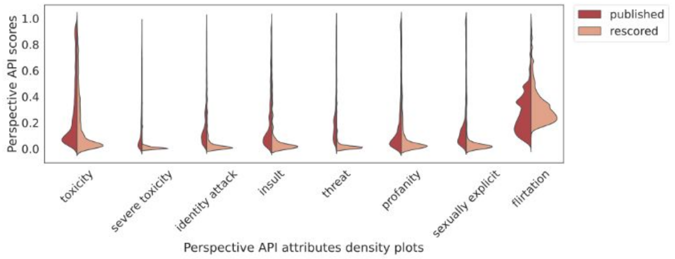


Figure 2: Rescored (Feb. 2023) and published (Sept. 2020) Perspective API attributes distributions from the RTP's prompts.

legal frameworks

research ethics

changing platform access options / ToS

user expectations / privacy

Tensions

platforms as black boxes

data access

data sharing

interdisciplinarity

methods as black boxes

publishing practices

missing data

Different Conclusions When It Comes To Addressing Specific Challenges

E.g., in the context of research ethics:

- ? Big vs. small data
- ? Users as authors vs. users as research subjects
- ? Particularly vulnerable groups (e.g., activists) vs. professional / public accounts (e.g., politicians)
- ? Different practices in quoting from user accounts based on disciplinary requirements

Weller, Katrin, and Katharina E. Kinder-Kurlanda. 2017. "To Share or Not to Share?: Ethical Challenges in Sharing Social Media-based Research Data." In Internet Research Ethics for the Social Age, edited by Michael Zimmer, and Katharina E. Kinder-Kurlanda, 115-129. New York u.a.: Peter Lang.

Different Conclusions When It Comes To Addressing Specific Challenges

Or in the context of data sharing:

- balancing between following principles of good scientific practice and between respecting legal constraints
- Perceived ethical obligations *towards the scientific community*
- Not sharing data to protect users vs. sharing to include users

Weller, Katrin, and Katharina E. Kinder-Kurlanda. 2017. "To Share or Not to Share?: Ethical Challenges in Sharing Social Media-based Research Data." In Internet Research Ethics for the Social Age, edited by Michael Zimmer, and Katharina E. Kinder-Kurlanda, 115-129. New York u.a.: Peter Lang.

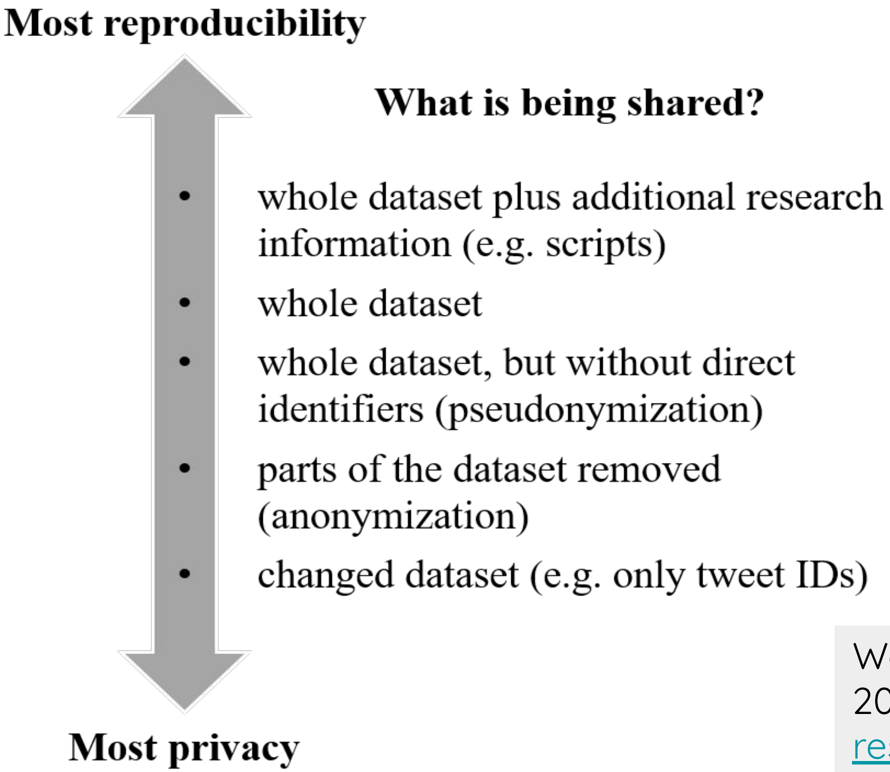
social media data aren‘t „ordinary“ research data

Perceived ethical obligations *towards social media users*

“It’s all public, it doesn’t belong to us, we don’t create the data, we don’t evoke it, I mean it’s natural. I don’t think you have the right to really keep other people from it, no.”

“We share datasets with everybody, actually. We don’t feel we own that.”

how much should I share?

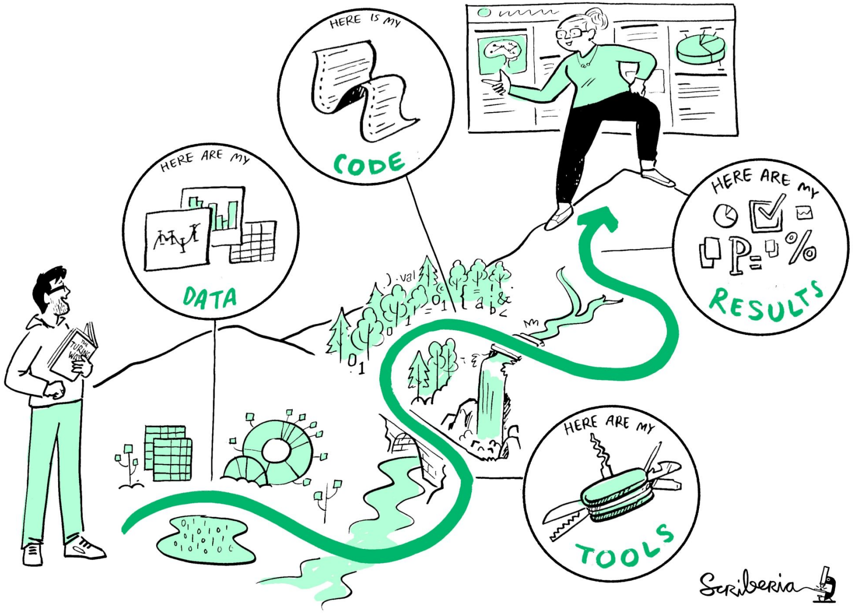


Weller, Katrin, and Katharina E. Kinder-Kurlanda. 2016. "A manifesto for data sharing in social media research." In Proceedings of the 8th ACM Conference on Web Science (WebSci '16), 166-172. New York: ACM.

How can you ensure your results are reproducible?

- Reproducible workflow from the very onset
  - Version control
  - Related: back up stuff regularly
  - Clear data management plan
  - Documentation

- When using social media data
  - Make sure the servers where your data is stored is secure



https://book.the-turing-way.org/reproducible-research/reproducible-research

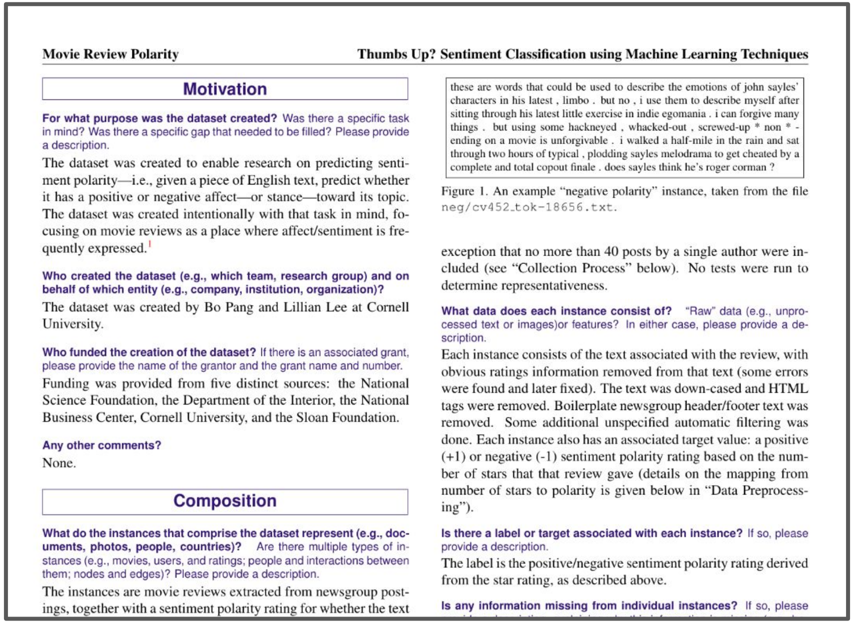
How can you ensure your results are reproducible?

- When using black-box software
  - Save everything! For example, save the labels and not just the final analysis
  - Make a note of when the black-box software was used

- Training and testing models
  - Functions are your friends; make everything as modular as possible
  - Clear definition of inputs and outputs
  - Save the models
  - Save the model outputs!

Getting ready to publish your data and materials

- Make your data as accessible as possible
  - For sensitive cases, ensure privacy and anonymity
  - Consider uploading to data archives like Zenodo or Harvard Data Archive instead of github
  - Include a license: https://choosealicense.com/licenses/
  - For particularly sensitive cases, consider gated access
  - For newly created data resources (e.g., a newly annotated dataset) add a data sheet or data statement



Movie Review Polarity — **Thumbs Up? Sentiment Classification using Machine Learning Techniques**

Motivation

For what purpose was the dataset created? Was there a specific task in mind? Was there a specific gap that needed to be filled? Please provide a description.

The dataset was created to enable research on predicting sentiment polarity—i.e., given a piece of English text, predict whether it has a positive or negative affect—or stance—toward its topic. The dataset was created intentionally with that task in mind, focusing on movie reviews as a place where affect/sentiment is frequently expressed.[1]

Who created the dataset (e.g., which team, research group) and on behalf of which entity (e.g., company, institution, organization)?

The dataset was created by Bo Pang and Lillian Lee at Cornell University.

Who funded the creation of the dataset? If there is an associated grant, please provide the name of the grantor and the grant name and number.

Funding was provided from five distinct sources: the National Science Foundation, the Department of the Interior, the National Business Center, Cornell University, and the Sloan Foundation.

Any other comments?

None.

Composition

What do the instances that comprise the dataset represent (e.g., documents, photos, people, countries)? Are there multiple types of instances (e.g., movies, users, and ratings; people and interactions between them; nodes and edges)? Please provide a description.

The instances are movie reviews extracted from newsgroup postings, together with a sentiment polarity rating for whether the text

these are words that could be used to describe the emotions of john sayles' characters in his latest , limbo . but no , i use them to describe myself after sitting through his latest little exercise in indie egomania . i can forgive many things . but using some hackneyed , whacked-out , screwed-up * non * - ending on a movie is unforgivable . i walked a half-mile in the rain and sat through two hours of typical , plodding sayles melodrama to get cheated by a complete and total copout finale . does sayles think he's roger corman ?

Figure 1. An example "negative polarity" instance, taken from the file `neg/cv452_tok-18656.txt`.

exception that no more than 40 posts by a single author were included (see "Collection Process" below). No tests were run to determine representativeness.

What data does each instance consist of? "Raw" data (e.g., unprocessed text or images)or features? In either case, please provide a description.

Each instance consists of the text associated with the review, with obvious ratings information removed from that text (some errors were found and later fixed). The text was down-cased and HTML tags were removed. Boilerplate newsgroup header/footer text was removed. Some additional unspecified automatic filtering was done. Each instance also has an associated target value: a positive (+1) or negative (-1) sentiment polarity rating based on the number of stars that that review gave (details on the mapping from number of stars to polarity is given below in "Data Preprocessing").

Is there a label or target associated with each instance? If so, please provide a description.

The label is the positive/negative sentiment polarity rating derived from the star rating, as described above.

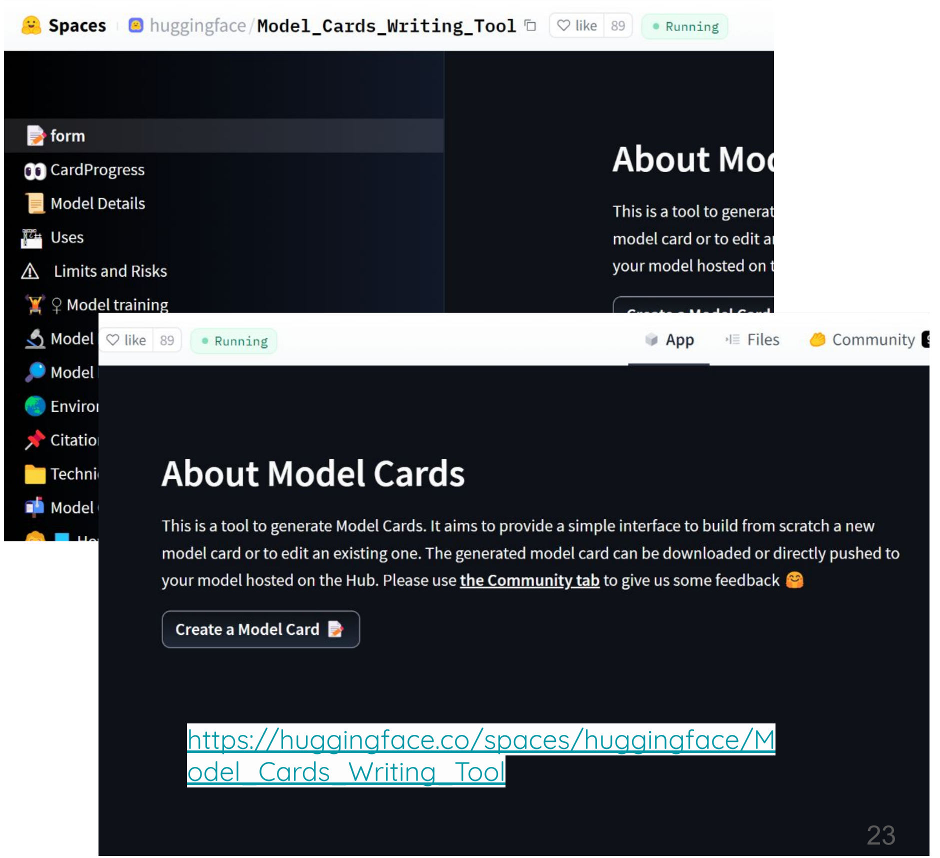
Is any information missing from individual instances? If so, please

[Datasheets for Datasets]()

Getting ready to publish your data and materials

- Respect data licensing agreements when using secondary dataset sources
    - Point to existing datasets rather than uploading a copy of it
    - Clearly describe how to access the data and how to preprocess it
    - Even better: upload the preprocessing script

- Code Documentation
    - Model cards



https://huggingface.co/spaces/huggingface/Model_Cards_Writing_Tool

Project Information

- Midterm presentation
- Final Report

Additional details are in the grading rubric:
https://github.com/dess-mannheim/SMDA/blob/main/grading_rubric.md

Midterm Presentation [15 minutes for each team, 15 points]

- Brief research question and motivation [2 points]
  - Include a slide on background — what research papers are most connected to your research question
- Dataset overview [4 points]
- Analysis overview [4 points]
- Preliminary results [4 points]
- Planned next steps [1 points]

Dataset Overview

- Include details on how you obtained the data
  - API, web scraping, etc
  - During which time period
  - With which search strategy (keywords, individual accounts, etc)
    - If list of keywords or accounts are too long, please include a few examples and have the full list in backup slides

- Summary statistics
  - Data size —> # posts, # users, engagement (# likes / favorites)
  - Distributions of your data, not just average statistics

- Data sampling and cleaning strategy

- For the report: comment on
  - why you think your dataset is suitable for your use cases
  - Which groups your data represents

Examples of Data Visualization in Social Media Research

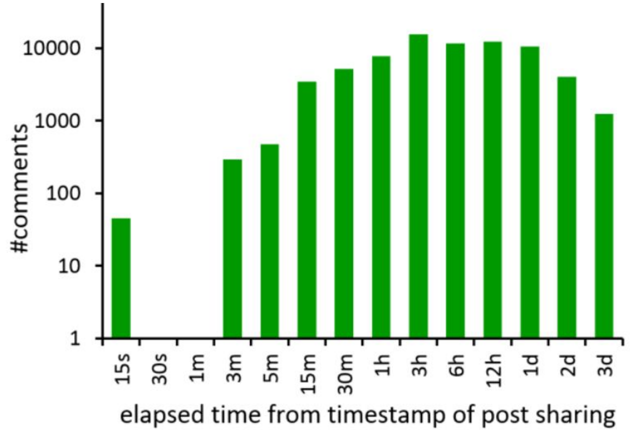


Figure 2. Comment distribution over time. Each bin shows the number of comments shared within a certain time interval from the timestamp of post.

De Choudhury, Munmun, and Sushovan De. "[Mental health discourse on reddit: Self-disclosure, social support, and anonymity.](#)" *Proceedings of the international AAAI conference on web and social media*. Vol. 8. No. 1. 2014.

Mental Health Discourse on reddit:
Self-Disclosure, Social Support, and Anonymity

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Abstract

Social media is continually emerging as a platform of information exchange around health challenges. We study mental health discourse on the popular social media: reddit. Building on findings about health information seeking and sharing practices in online forums, and social media like Twitter, we address three research challenges. First, we present a characterization of self-disclosure in mental illness communities on reddit. We observe individuals discussing a variety of concerns ranging from the daily grind to specific queries about diagnosis and treatment. Second, we build a

online fora, these social systems are more holistic, in the sense that millions of people use them to post about the mundane goings on of their lives. Additionally, unlike many online communities, most social media sites have a personal permanent identity associated with user profiles. Consequently, these platforms provide a rich ecosystem to study the variety of self-disclosure, social support, and disinhibition that engender health related discourse.

Specifically, we examine a highly popular social news and entertainment media: reddit (*http://www.reddit.com/*). n interesting online social system that has the f a forum: it allows sharing blurbs of text and osts that invite votes and commentary. At the it is often used as a social feed of information d from people's contacts and audiences. Another aspect of reddit is its "throwaway" accounts. emporary identities often used as an "anonymity iscuss uninhibited feelings, sensitive information, unacceptable thoughts momentarily; information considered unsuitable for the mainstream. e unique characteristics of reddit, in this paper rested in studying the nature of discourse around ant health challenge of mental illness. Mental

Posts per day	247.1 (±41.3)
Average posts per user	0.75 (±2.21)
Mean, Median length of posts (words)	96.4, 64
Average upvotes per post	8.37 (±23.2)
Average downvotes per post	1.26 (±3.71)
Average comments per post	5.03 (±7.52)
Mean, Median comment length (words)	27.03, 16
Average comments per user	2.69 (±8.49)
Average upvotes per comment	2.39 (±4.16)
Average downvotes per comment	2.06 (±3.58)
Total number of posts	72,236

Table 1. Basic descriptive statistics of the reddit dataset.

General tips, especially for your reports and presentations

Bakhshi, Saeideh, David A. Shamma, and Eric Gilbert. ["Faces engage us: Photos with faces attract more likes and comments on instagram."]() Proceedings of the SIGCHI conference on human factors in computing systems. 2014.

- Descriptive statistics:
  - Tables or figures showing summary of your data
  - especially the variables you are mainly studying (distributions, min, max, averages)

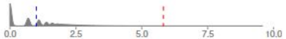
Type	Variable	Description	Distribution
Engagement	likes*	Number of likes on each photo.	
	comments*	Number of comments on each photo.	
Audience & Activity	followers*	Number of users who follow the photo's owner.	
	photos*	Number of photos shared by photo's owner.	
Faces	has face	1 if the photo contains a face, 0 otherwise.	
	has face $<$ 18 years old	1 if there is at least one face younger than 18, 0 otherwise.	
	has face $\in$ [18, 35] years old	1 if there is at least one face with age between 18 and 35, 0 otherwise.	
	has face $>$ 35 years old	1 if there is at least one face older than 35, 0 otherwise.	
	has female face	1 if there is at least one female face in the photo, 0 otherwise.	
	has male face	1 if there is at least one male face in the photo, 0 otherwise.	

Table 1. Distributions of quantitative and binary variables used in this paper. Variables marked with '*' are log-transformed. The red and blue lines identify mean and median of the distribution, respectively. Orange refers to 1's in the bar graphs. The engagement variables are our dependent measures. Audience and activity variables are used as controls, and faces variables are the focal point of this study.

Analysis Overview

- Clear overview of methods with references to them
- Justification for why these methods
- Validation and which metrics you use for validation
    - For the midterm presentation: it’s fine if you state your validation plan
    - For the report: you should have the validation and state the limits of your approach

Results

- Use visualizations as much as possible
- For the midterm: focus on 1-2 main results that are most interesting
- Make sure plots are
    - Labeled (all axes, ticks, legends)
    - Legible
- Use colors and markers to differentiate different modes

General tips, especially for your reports and presentations

- Data visualization:
  - All axes should have labels
  - Captions should be self-contained
  - The text in the figure should be as big as the text in your paper
  - Using colors to convey information is nice, but also good to use markers or textures for this
  - Also keep in mind accessibility:
  - https://towardsdatascience.com/two-simple-steps-to-create-colorblind-friendly-data-visualizations-2ed781a167ec

Report Logistics

- Due on **7.07.2026** at 23:59 on ILIAS, optional pre-submission via email (to me, Georg, and Maxi) on 22.06.2026
    - If you pre-submit, we will send you our preliminary feedback on critical issues that you should address by 29.06
    - You can also send the report later, but we cannot guarantee that you will get feedback in time before the final submission

Midterm Logistics

- Ideally: all 6 teams are present on both days
    - Who can only attend one of the sessions? Which one?
- Submit slides on ILIAS by next Wednesday, 27.05.2026, 10 AM, irrespective of if you present on Wednesday or Thursday
- All presentations will be done via my laptop, followed by Q&A

Report Logistics

- Use the ACL template: https://github.com/acl-org/acl-style-files

- Reports should be 4-5 pages long (main content), with unlimited space for references and appendices.

- Ideally the main paper should have 1-2 interesting results while rest of the details are in the appendix
  - Be sure to reference all materials in the appendix in the main part
  - Be sure all figures and tables are referenced in the text
  - Be sure that figures and tables have detailed, self-contained captions, i.e., that anyone who reads just the caption and the table/figure can understand it without having to see the text

Report Logistics

- Code and Dataset: Your codebase and dataset (or a sample of it if it's too large) should be available in a Github repository with sufficient documentation to ensure reproducibility
  - Put a link to the repository in the report
  - For private repositories, please add me, Maxi, and Georg

- Dataset Anonymization
  - Check guidelines if you are actually allowed to share the data
  - If not, try to anonymize it or share a sample
  - If you choose not to share the full, raw data, be sure to have a justification in the repository with clear instructions on how the data can be recreated

AI Usage in Presentations and Report

- You are not forbidden from using AI systems for your work

- But you are 100% responsible for all the contents of it

- If you have errors in your work, **<u>you</u>** will be held accountable for them
    - E.g., if you have hallucinated or incorrect citations, that is considered academic dishonesty and will lead to penalties
    - E.g., if your code doesn't run, that will lead to penalties

- You should also be fully prepared to answer questions on what you did
    - If we feel that you have used AI without understanding what it did, we reserve the right to ask you questions about your work and re-evaluate the project based on your answers

Lecture Next Week

- Course summary and outlook

- Readings:
    - Lazer, David MJ, et al. "Computational social science: Obstacles and opportunities." Science 369.6507 (2020): 1060-1062.
    - Sen, Indira, et al. "A total error framework for digital traces of human behavior on online platforms." Public Opinion Quarterly 85.S1 (2021): 399-422.

- Next week's lecture and midterm presentations are the last chance to get class participation points

Questions?

Additional Resources

- The Turing Way's Guide to Reproducible Research: https://book.the-turing-way.org/reproducible-research/reproducible-research
- Course of effective research workflows: https://www.brendanmichaelprice.com/workflow/ focusing on many details like
  - Readable code
  - Collaborating well
- Schoch et al'2023 "Computational Reproducibility In Computational Social Science"